Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Wireless Networking

Task:

❖ **1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.**

➢ **Aim**

- To identify three popular Wi-Fi standards and understand their key differences in speed and frequency bands.

➢ **Introduction**

- Wi-Fi is a wireless networking technology that allows devices such as laptops, smartphones, and tablets to connect to a network without using physical cables. Wi-Fi standards are developed by the IEEE (Institute of Electrical and Electronics Engineers) under the 802.11 family.
- Different Wi-Fi standards provide different data speeds, frequency bands, and network performance.

➢ Popular Wi-Fi Standards

No.	Wi-Fi Standard	Frequency Band	Maximum Theoretical Speed	Key Difference
1	**802.11n (Wi-Fi 4)**	2.4 GHz and 5 GHz	Up to 600 Mbps	Supports both 2.4 GHz and 5 GHz bands and offers better speed than older Wi-Fi standards.
2	**802.11ac (Wi-Fi 5)**	5 GHz	Up to about 6.9 Gbps	Operates in the 5 GHz band and supports higher data rates than 802.11n.
3	**802.11ax (Wi-Fi 6)**	2.4 GHz and 5 GHz	Up to about 9.6 Gbps	Provides higher theoretical speed and improved efficiency when many devices connect simultaneously.

➢ Explanation

➢ 1. IEEE 802.11n (Wi-Fi 4)

802.11n supports both 2.4 GHz and 5 GHz frequency bands. It introduced technologies such as MIMO (Multiple Input Multiple Output) to improve wireless performance and data speed.

➢ 2. IEEE 802.11ac (Wi-Fi 5)

802.11ac operates in the 5 GHz band and supports higher data rates through wider channels and improved MIMO technology. It is suitable for activities such as HD video streaming and online gaming.

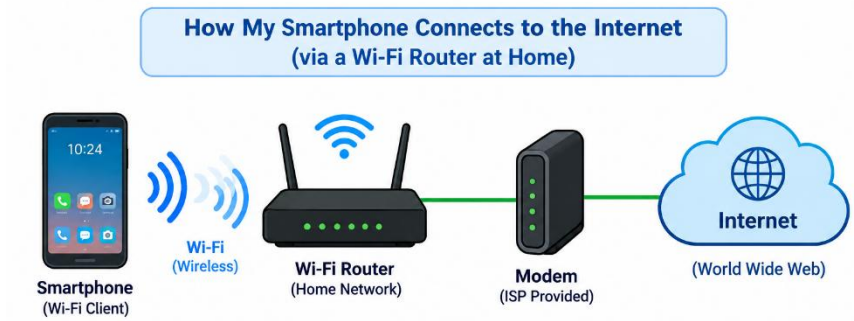
➢ **3. IEEE 802.11ax (Wi-Fi 6)**
802.11ax supports both 2.4 GHz and 5 GHz bands. It improves network efficiency in crowded environments using technologies such as OFDMA, allowing multiple devices to communicate more efficiently.

➢ **Result**
➢ Three popular Wi-Fi standards—802.11n, 802.11ac, and 802.11ax—were studied. Their frequency bands, theoretical maximum speeds, and key differences were identified.

❖ **2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.**

➢ **Aim**
- To illustrate how a smartphone connects to the Internet through a home Wi-Fi router and identify the wireless and wired parts of the network.

➢ **Diagram**



- **Smartphone → Wi-Fi Router → Modem → Internet**
- Label the connections as follows:
  - **Smartphone to Wi-Fi Router:** Wireless connection (Wi-Fi signals)
  - **Wi-Fi Router to Modem:** Wired connection (Ethernet cable)
  - **Modem to Internet:** Connection provided by the Internet Service Provider (ISP)

- **Working**
  1. The smartphone connects to the home Wi-Fi router using wireless radio signals.
  2. The Wi-Fi router forwards the smartphone's network traffic towards the modem.
  3. The modem connects the home network to the Internet Service Provider (ISP).
  4. The ISP provides access to the Internet.
  5. The requested data returns through the network to the smartphone.

- **Result**
- The connection between a smartphone and a home Wi-Fi router is wireless, while the connection between the router and modem is typically wired using an Ethernet cable.

❖ **3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Consider both signal strength and security when choosing.**

➢ **Aim**

- To scan available Wi-Fi networks, record their SSIDs, signal strengths, and security types, and select the most suitable network based on signal strength and security.

➢ **Introduction**

- Wi-Fi is a wireless networking technology that allows devices such as laptops and smartphones to connect to a network without physical cables.
- A Wi-Fi network has an SSID (Service Set Identifier), which is its network name. Signal strength indicates how strongly the device receives the wireless signal. The security type, such as WPA2 or WPA3, indicates the protection used to secure the wireless connection.
- Checking these details helps users select a reliable and secure Wi-Fi network.

➢ **Procedure**

1. Turn on Wi-Fi on the laptop or smartphone.
2. Open the available Wi-Fi networks list in the device settings.
3. Scan the available networks.

4. Record the SSID, signal strength, and security type of at least three networks.
5. Compare the networks based on their signal strength and security.
6. Select the most suitable network and explain the reason for the selection.

➢ **Observation Table**

Sr. No.	SSID (Network Name)	Signal Strength	Security Type
1	My Wi-Fi Network	Strong	WPA2-Personal
2	Office_WiFi	Medium	WPA3-Personal
3	Guest_WiFi	Weak	Open (No Password)

➢ **Comparison and Selection**

➢ The most suitable Wi-Fi network should provide both good signal strength and strong security.

- **Signal Strength:** A strong signal generally provides a more reliable connection and reduces the likelihood of connection problems.
- **Security:** WPA3 is generally preferred when supported. WPA2 with AES is also a widely used secure option. Open networks without password protection should be avoided for sensitive activities.

- **Final Selection:** I would choose the network that provides a strong signal and uses WPA3 security, or WPA2-AES if WPA3 is unavailable.

➢ **Result**

➢ Three available Wi-Fi networks were scanned and compared based on their SSIDs, signal strengths, and security types. The most suitable network can be selected by considering both connection quality and security.

❖ **4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.**

➢ **Introduction**

- SSID stands for **Service Set Identifier**. It is the name used to identify a Wi-Fi network. For example, when we open the Wi-Fi settings on a smartphone or laptop, names such as Home_WiFi or Office_WiFi are examples of SSIDs.
- SSID hiding is a feature in a Wi-Fi router that prevents the network name from being displayed in the normal list of available Wi-Fi networks. Users who want to connect to the network generally need to know its name and enter the connection details manually.

➢ **Advantage of SSID Hiding**

➢ **Reduced Visibility:** Hiding the SSID makes the Wi-Fi network less visible to ordinary users browsing the available Wi-Fi network list. This may discourage casual users from attempting to connect to it.

- **Disadvantage of SSID Hiding**
- **It Does Not Provide Strong Security:** Hiding the SSID does not prevent a determined person from discovering the network name using suitable wireless scanning tools. It can also make connecting new devices less convenient because users must enter the network name manually.